

Deep Learning Approaches for Vietnamese Sign Language Recognition

This paper presents a comprehensive survey of deep learning architectures for automatic recognition of Vietnamese Sign Language (VSL) from video sequences. We evaluate CNN, LSTM, and Transformer-based models on VSL-5K, a new benchmark dataset of 5,000 videos spanning 200 VSL gestures. Our proposed Hybrid Vision-Transformer (HVT) achieves 94.7% accuracy, outperforming existing methods by 3.2pp. We release our dataset, models, and evaluation code as open-source at <https://github.com/vnsl/hvt>.

Keywords: Sign Language, Deep Learning, Computer Vision, Transformer, Gesture Recognition

1. INTRODUCTION

Sign Language Recognition (SLR) is a critical task in assistive AI. Vietnamese Sign Language (VSL) presents unique challenges: high intra-signer variability, regional dialect influence, and simultaneous multi-channel expression (hands + facial cues). Prior work has primarily focused on ASL and BSL, leaving VSL understudied.

This paper contributes: (1) VSL-5K — a 5,000-video benchmark across 200 gestures from 50 native signers. (2) HVT — a Hybrid Vision-Transformer architecture. (3) An ablation study quantifying the contribution of each model component.

2. RELATED WORK

Early SLR relied on video sensors (Kong & Ranganath, 2018). CNNs became dominant for isolated sign recognition (Koller et al., 2019). LSTMs modeled temporal sequences (Cavazza et al., 2020), while Transformers captured long-range dependencies (Ali et al., 2021). Despite these advances, no prior work addresses the multimodal nature of VSL at scale.

3. METHODOLOGY

3.1. Pipeline Overview. (1) CNN spatial encoder (ResNet-50) extracts per-frame visual features. (2) Vision Transformer applies multi-head self-attention across spatial tokens. (3) Temporal Transformer aggregates frame-level representations across T time steps. We use a weighted cross-entropy loss and AdamW optimizer with cosine annealing LR.

3.1. [Swin Transformer](#) overview, top-1 accuracy, top-5 accuracy, and per-frame latency (ms). Table 1 summarizes results on the VSL-5K test set.

Transformer models spatial attention; the Temporal Transformer aggregates across T frames.

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Table 1: Comparison of the SUR Method on the SSF-ACT dataset. The Temporal Transformer is critical for capturing rapidly changing gesture dynamics, while the Vision Transformer distinguishes visually similar hand configurations. Figure 2 (next page) shows the per-class accuracy breakdown. Errors concentrate among visually similar gestures differing primarily in index finger direction.

CONCLUSION

We propose **ViSL-Transformer**, a hybrid vision-transformer for Vietnamese Sign Language recognition, achieving 94.7% accuracy on VSL-5K. Key findings: (1) the temporal Transformer captures motion dynamics; (2) spatial attention distinguishes similar hand shapes; (3) class-weighted loss handles the long-tail gesture distribution. Future work includes extending to continuous sign language recognition (CSLR) with CTC loss, incorporating facial landmark features for expression modeling, and mobile deployment for real-time inference.

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